

Jahid Md Hasan

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SUMMARY

Machine Learning Engineer building scalable ML and GenAI systems. Experienced in developing RAG pipelines and multi-step Agentic AI workflows using Transformers and LLM orchestration frameworks. Designed modular, production-ready ML architectures across finance, healthcare, and cybersecurity domains with integrated data validation, drift monitoring, and explainability. Skilled in deploying models on AWS using CI/CD-driven workflows to deliver reliable, real-world AI systems.

WORK EXPERIENCE

Machine Learning Intern - Unified Mentor

June 2025 - Sep 2025

- Developed and deployed multiple end-to-end machine learning pipelines across healthcare, finance, and accessibility domains, applying advanced techniques in EDA, feature engineering, and model tuning.
- Built a fraud detection system using temporal transaction data and domain-specific simulation rules, leveraging feature engineering, imbalanced classification techniques, and ensemble models (XGBoost, Random Forests).
- Engineered a healthcare diagnostic model for heart disease prediction, incorporating explainable AI tools (SHAP/LIME) and performance metrics (ROC-AUC, precision-recall) for clinical interpretability.

Data Science Intern - InLighnX Global

Oct 2025 - Feb 2026

- Built an Air Quality Prediction System using Random Forest, Prophet, and LSTM, achieving $R^2 = 0.9968$, with advanced feature engineering, automated preprocessing, and interactive forecasting dashboard deployment.
- Developed a Voice-Activated AI Chatbot integrating speech recognition, text-to-speech, NLP-based command routing, API integrations, and persistent JSON storage for modular and event-driven automation.
- Applied end-to-end ML lifecycle practices including data cleaning, feature engineering, model evaluation, serialization (Joblib/Keras), and deployment-ready packaging.

PROJECTS

Network Security for Phishing Data

[View in GitHub](#)

- Designed and implemented a **production-grade modular ML pipeline** for network intrusion detection, covering ETL, Data Ingestion, Data Validation, and Data Transformation with artifact-driven architecture.
- Built a **schema validation and data drift detection system** using KS-test (scipy) to ensure training data consistency and prevent silent model degradation in production environments.
- Developed a **configuration-driven, timestamped artifact system** with custom logging and exception handling, enabling reproducibility, observability, and scalable ML workflows.
- Integrated **MongoDB Atlas as centralized data source** and implemented a feature store mechanism with automated train-test splitting and preprocessing pipeline (KNN Imputer), saving transformed data as NumPy artifacts for downstream model training.

- Developed an end-to-end Neural Machine Translation system using the Transformer architecture, leveraging self-attention mechanisms for improved linguistic context.
- Optimized model performance for English-to-Hindi translation, focusing on tokenization and hyperparameter tuning to achieve higher BLEU scores.
- Integrated the model into a web interface for real-time translation, ensuring low-latency inference.

CERTIFICATIONS

IBM Data Science	View Credentials
GenAI for Data Scientists	View Credentials
RAG for Generative AI Applications	View Credentials

EDUCATION

Amity University – Noida (Online)	Sep 2023 – Jan 2026
Master of Computer Applications (MCA) – Specialization in AR/VR & Intelligent Systems	
Assam University – Silchar (India)	Aug 2019 – Nov 2022
Bachelor of Science (BSc.) – Computer Science Honours	

SKILLS

Programming :	Python, C++, Java, JavaScript, SQL
Machine Learning & Deep Learning :	PyTorch, TensorFlow, Scikit-learn, XGBoost, LSTM, Transformer Architectures
GenAI & LLM Systems :	Hugging Face, LangChain, RAG Pipelines, Agentic Workflows, Prompt Engineering
MLOps & Deployment :	AWS (EC2, S3, Lambda), Docker, FastAPI, CI/CD , Model Serialization(Joblib/Keras)
Data & Infrastructure :	MongoDB Atlas, ETL Pipelines, Data Validation, Drift Detection, Feature Engineering